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Spatial distribution of ion emission in gas-puff z-pinches and dense plasma foci

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Abstract

Mega-ampere dense plasma foci and deuterium gas-puff z-pinches can accelerate deuterons to multi-MeV energies. Diagnostic measurements of the properties of these ions provide information about ion acceleration in z-pinch plasmas. In particular, the results from ion pinhole cameras seem to be useful for the discussion of ion acceleration mechanisms. Recently, we have used various configurations of ion pinhole cameras in deuterium gas-puff experiments on the GIT-12 generator at the Institute of High Current Electronics in Tomsk and on the HAWK generator at the US Naval Research Laboratory in Washington. The stack of radiochromic films and CR-39 solid-state nuclear track detectors recorded deuterons with energies up to 30 MeV. From our ion diagnostics, we obtained the spatial distribution of the ion source and the ion-beam divergence during the ion emission. This ion-beam divergence was found to decrease with increasing deuteron energy. At 20 MeV, the divergence of each of the individual micro-beams that composed the ion source was on the order of 10 mrad. The deflection of each micro-beam due to the azimuthal magnetic and/or radial electric fields resulted in radial stripes observed by the beam-profile detectors. By analyzing the ion pinhole images, we found that the deuterons were emitted both from a central spot and from a ring-shaped region with a rather large diameter, on the order of 1 cm. The origin and particular diameter of this ring is attributed to the geometry of the electrodes and to the distribution of the current density before the disruption.

Keywords: z-pinch, dense plasma focus, multi-MeV deuterons, acceleration mechanism, plasma diagnostics

(Some figures may appear in colour only in the online journal)

1. Introduction

Deuterium z-pinches and dense plasma foci (DPFs) have been researched as efficient sources of fast neutrons [1–5]. A large number of neutrons originating from the $D(d,n)^3\text{He}$ reaction

were observed in linear deuterium pinches in the 1950s [6–8]. However, these neutrons were not of thermonuclear origin (see [9, 10]), i.e. they were not produced by the fusion of deuterons accelerated by elastic collisions in high-temperature plasmas. At the very beginning of fusion research, the dominant fraction of neutrons observed in z-pinches was explained as arising from deuterons that were accelerated to fusion-inducing energies by

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transient electric fields [6, 8]. Since then, many experiments with various diagnostic tools have been performed [1, 11, 12], and many hypotheses have been proposed to explain the acceleration of deuterons in z-pinches and DPF [13, 14]. Recently, state-of-the-art numerical codes have been used to solve this very old problem in plasma physics [15–17]. Despite these efforts, the exact acceleration mechanism has still not been resolved [14].

One of the unexplained results was the observation of ring-like structures in ion pinhole images. We suppose that a better understanding of the intriguing ion pinhole images and the correct answer to the question of where ions are accelerated could help with identification of the acceleration mechanism. Interestingly, characteristic annuli were observed by ion pinhole cameras both in dense-plasma foci [18–25] and in gas-puff z-pinches [26–28]. This is an important result, since it indicates that the acceleration of high-energy ions can take place in an analogous way in both devices. Taking into account other similarities, such as the correlation of ion and neutron emission with an $m = 0$ disruption [14], the acceleration mechanism seems to be quite general and might occur in various current-carrying plasmas. From this point of view, the explanation of the ring-like structures in pinhole images is of great importance.

40 years ago, Filippov *et al* [18] suggested that the annuli were created by deuterons which were emitted from an on-axis source and then bent towards the pinhole by the magnetic fields of the current sheath. The second hypothesis was proposed by Bertalot with his colleagues [19], who claimed that ions originated from ‘many point sources localized off-axis, possibly on the surface of the plasma region’. In other words, Bertalot was of the opinion that the rings in the pinhole images reflected the spatial distribution of the original ion source. Both research groups did not prove their hypotheses. Unambiguous interpretation of the ring-shaped patterns has not been provided, even in other DPF experiments with single or multi-pinhole cameras [20–25]. The main difficulty was that the study of ion emission was inevitably connected to the knowledge of the electric and magnetic fields in z-pinches or DPF. Since the effect of the magnetic fields on the ion trajectories was not known, the experimental results were not sufficient to distinguish even between the two completely different hypotheses mentioned above. To solve this issue, it is necessary to obtain comprehensive information about the ion emission. For this purpose, we have carried out deuterium gas-puff experiments on the GIT-12 and HAWK microsecond generators at 3 MA and 0.7 MA current, respectively. GIT-12 is unique in the acceleration of deuterons to unprecedented high energies up to 40 MeV [26, 27, 29, 30]. The high deuteron energies enable us to diagnose the ions by several novel (to z-pinches) techniques (nuclear methods [30], radiochromic film (RCF) stack spectroscopy [31, 32], etc) and, therefore, to obtain comprehensive information about the spatial, spectral, and temporal properties of the accelerated ions.

In this paper, we will present results from various configurations of ion pinhole cameras. Section 2 describes the z-pinch load used in our experiment on GIT-12. Section 3 introduces characteristic results from the ion pinhole cameras. Ion pinhole images with spectral resolution are presented in section 4. Section 5 brings forward results from a multi-

pinhole camera which provided some conclusions about the anisotropy of the ion emission. These conclusions about the anisotropy were confirmed by three-pinhole cameras with beam-profile detectors which are described in section 6. Section 7 correlates pinhole images with neutron yields. Section 8 deals with the origin of ion rings of a particular diameter and presents the results from a pinhole camera on the HAWK generator. Finally, section 9 summarizes the most important points of this paper with respect to ion acceleration mechanisms.

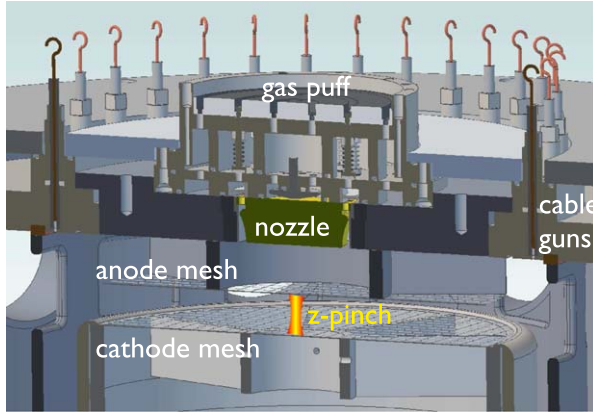
2. Experimental set-up and diagnostics on GIT-12 and HAWK

2.1. GIT-12 generator at the Institute of High Current Electronics in Tomsk

Z-pinch experiments with deuterium gas puffs have been carried out on the GIT-12 generator at a 3 MA current and microsecond rise-time [33]. In order to form a homogeneous, uniformly conducting layer at a large initial radius, an inner 8 cm diameter deuterium gas puff was surrounded by an outer hollow cylindrical plasma shell [5, 29]. The plasma shell, consisting of hydrogen and carbon ions, was formed at a diameter of 35 cm by 48 cable guns (see figure 1). When the plasma guns were triggered 1.7–1.8 μs before the onset of the main current, the linear mass of the plasma shell was estimated to be $5 \mu\text{g cm}^{-1}$.

As shown in figure 1, the nozzle was placed on the anode side 3.0–3.6 cm from the anode mesh. The anode–cathode gap varied between 2.0 and 2.8 cm for most of the shots. The diameter of the cathode was 32 cm, and the inner diameter of the return-current conductor was approximately 37 cm. Both the anode and cathode were formed by a stainless-steel mesh with a transparency of about 70%. The optimal time delay between the valve opening and the triggering of the generator varied between 250 and 330 μs . Due to the relatively long injection time and multiple reflections of the gas from the mesh electrodes, the gas was spread over a large area and the gas-density profile was expected to be quite smooth (see also the 2D ANSYS FLUENT CFD simulation in figure 1 in [26]). The total linear mass of deuterium in the single shell (annular) gas puffs was usually between 80 and 120 $\mu\text{g cm}^{-1}$. During implosion, it is assumed that some mass is lost through the meshes.

The initial parameters mentioned above led to a deuterium gas-puff implosion that lasted about 700 ns (measured from the time at which the load current reached 100 kA). Soft x-ray images showed a DPF-like (smooth and curved) imploding layer that was stable until it reached a radius of 4 mm (see figure 2(a)). The z-pinch column stagnated at a current of 2.7 MA. During stagnation, $m = 0$ instabilities became more pronounced near the anode (see figure 2(b)). At $t = 0$, the disruption of an $m = 0$ neck occurred [26], and a bright spot was detected by the radial streak camera between 0 and 10 ns (see figure 2(d)). The moment of the z-pinch disruption corresponded to the time of emission of high



(a) Schematic diagram of the experimental arrangement with electrodes, gas-puff hardware, and cable guns.



(b) Photograph of the experimental chamber with 48 cable guns.

Figure 1. Experimental arrangement of the electrode system, gas-puff hardware, and cable guns on GIT-12.

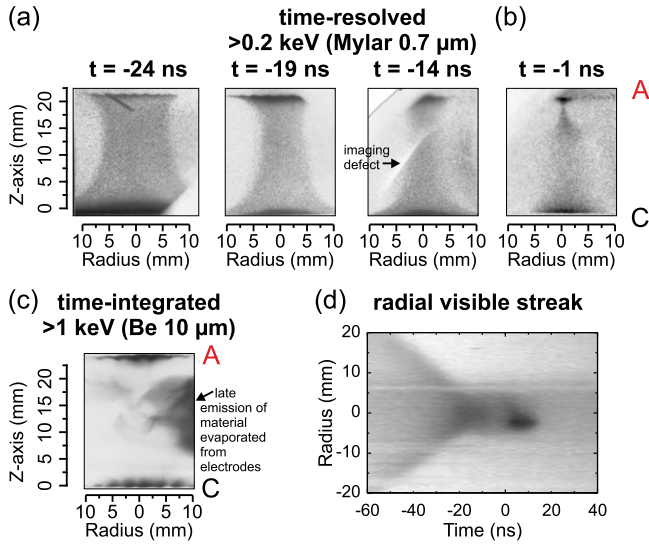


Figure 2. Time-resolved soft x-ray images of a deuterium gas-puff z-pinch on shots (a) 1771 and (b) 1764. The anode is at the top and the cathode is at the bottom. Note: The x-ray pinhole framing camera did not have a full view of the anode mesh at the z-pinch axis. (c) Time-integrated soft x-ray image and (d) radial streak-camera image of a deuterium gas-puff z-pinch on shot 1771. The time $t = 0$ corresponds to the sharp onset of a γ -ray pulse and a bright spot in the streak camera image. Shot 1771 with $(2.0 \pm 0.4) \times 10^{12}$ neutrons.

energy (>2 MeV) bremsstrahlung radiation and the main neutron pulse [5, 26]. The average neutron yield was 2×10^{12} per shot.

The implosion, stagnation, and disruption of the deuterium gas-puff z-pinch were observed by a comprehensive set of diagnostics. The electrical, optical, x-ray, gamma, and neutron detectors were described in [5], and ion diagnostics were presented in [26, 30]. Several new ion detectors will be introduced later in this article.

2.2. HAWK generator at the Naval Research Laboratory in Washington

HAWK is a high-inductance (607 nH) generator located at the US Naval Research Laboratory in Washington [34]. The pulsed-power architecture of the HAWK generator is similar to that of GIT-12. At an 80 kV charging voltage, HAWK stores an energy of 0.22 MJ and delivers a 0.64 MV, 0.7 MA pulse to a load within a $1.2 \mu\text{s}$ rise time. Since 2017, the HAWK generator has been used to drive a DPF. The arrangement of the electrode system is presented in figure 10(a) in section 8. The anode and cathode diameters were 10.5 and 17.3 cm, respectively. In the DPF configuration on HAWK, 3 Marshall guns injected a deuterium plasma shell radially between coaxial electrodes. The total plasma shell mass was $1 \mu\text{g cm}^{-1}$. The additional mass of neutral deuterium gas was puffed by an on-axis, 3.8 or 4.45 cm diameter valve placed inside the central anode [35]. In a recent experimental campaign with reproducible high neutron yields of about 3×10^{10} , the mass per unit length was about $30 \mu\text{g cm}^{-1}$. Unlike the experimental set-up on GIT-12, there was no cathode mesh on HAWK. Therefore the initial set-up of the HAWK load might be considered similar to the radial implosion phase of a dense plasma focus where the length of the imploding plasma column is not fixed [36]. A detailed description of the experimental set-up is the subject of an upcoming paper [37]. Preliminary results were published in [27, 38]. Recent data from an ion pinhole camera on HAWK will be presented at the end of this paper in section 8. In the following sections 3–7, we will present the results from ion diagnostics on GIT-12.

3. Characteristic annuli in ion pinhole images

Multi-MeV deuterons were observed in the GIT-12 z-pinch experiments first in 2013 [29]. Since then, a large number of

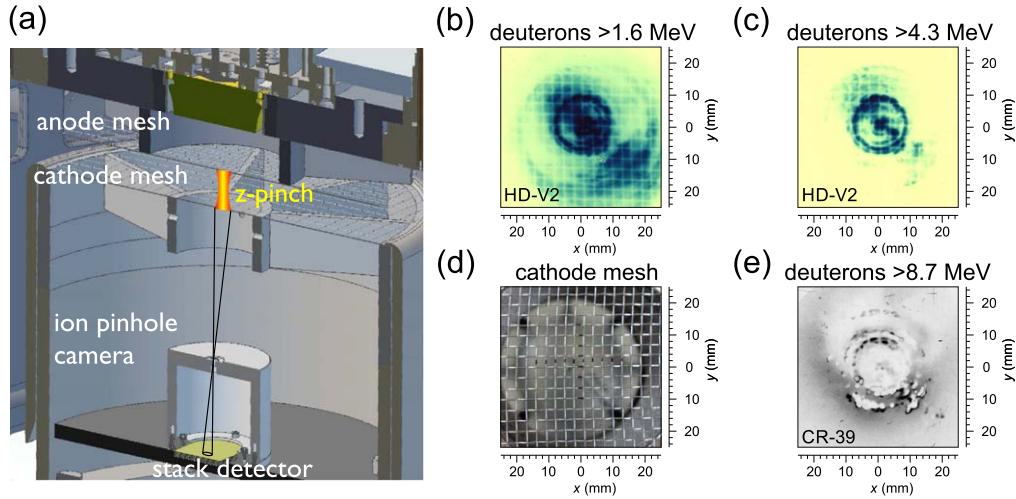


Figure 3. Measurement of the spatial distribution of the ion source. (a) Schematic showing the arrangement of the axial ion pinhole camera with a magnification of 0.6, a pinhole diameter of 0.5 mm, and a spatial resolution of 1.3 mm. (b)–(c) Axial ion pinhole images recorded by the first and second HD-V2 GafChromic films in the stack. (d) On-axis view of the cathode mesh. (e) Axial ion pinhole image recorded by a CR-39 TasTrak detector. Shot no. 1750 with $(1.2 \pm 0.3) \times 10^{12}$ neutrons. Spatial scales correspond to the plane of the cathode mesh. The detector darkness is proportional to the ion flux.

ion detectors have been used to characterize the ion emission and to investigate the ion acceleration mechanism. Among these diagnostic measurements, a characteristic structure was observed in the ion pinhole images obtained. Although the origin of this structure was not clear, we supposed it was connected to the ion acceleration mechanism and that an ion pinhole camera would be helpful in identifying that mechanism. In most of the shots, we placed the ion pinhole camera on the z-pinch axis below the cathode mesh. The arrangement of the experimental set-up and typical results are shown in figure 3. The ion pinhole camera used a stack consisting of absorbers, RCFs, and solid-state nuclear track detectors. Deuterons with different energies were stopped in different layers. Ions with lower energies were stopped at the beginning of the stack, whereas deuterons with higher energies penetrated the first detectors and got stopped in the deeper layers of the stack. The minimum deuteron energy required for each layer was evaluated by the stopping and range of ions in matter (SRIM) code [39]. This minimum energy value is shown along with each ion image presented in this paper. The signal in a particular ion detector layer within the stack was mainly due to ions near their Bragg peak. But more energetic deuterons that largely passed through the layer also contribute to the total dose absorbed in that layer. As ion detectors, we used FWT-60 RCFs, HD-V2 and EBT-3 GafChromic films, and CR-39 TasTrak solid-state nuclear track detectors. Each of these detectors has different properties. Therefore it proves beneficial to use them simultaneously to obtain complementary information. The advantage of RCFs is that they are relatively thin and can provide a high dynamic range for radiation dose detection, namely 0.1–20 Gy (EBT-3), 10–1000 Gy (HD-V2), and 0.5–200 kGy (FWT-60). As for CR-39 detectors, an important property is that they are insensitive to x-rays and electrons [40]. In addition to that, CR-39 detectors enable us to detect individual ions and distinguish different ion species. Therefore, we used CR-39

detector layers mainly at the bottom end of each detector stack (see e.g. figure 4(a)). In our experiment, CR-39 detectors were usually etched for 2 h in 30% KOH at 70 °C. At higher ion fluences detected by CR-39, we needed to be mindful of saturation effects and consequent artifacts (see [41] and compare figures 3(c) with (e)). The data from the CR-39 detectors, particularly the agreement between the signals on CR-39 and on adjacent RCFs, indicated that we could safely neglect the contribution of x-rays and electrons to the doses measured in HD-V2 GafChromic films fielded in ion pinhole cameras placed below the cathode.

As shown in figures 3(b)–(c), the characteristic feature recorded by the pinhole images was ion emission from both a central spot and ring-like structures of a relatively large diameter. At low deuteron energies, i.e. below 3 MeV, these rings were blurred. The unperturbed shadow of the cathode mesh is apparent in the pinhole images, indicating that the ions are transported ballistically below the cathode mesh. The images in figure 3 then represent the distribution of ions at the plane of the cathode mesh whose direction of travel and energy allow them to pass through the pinhole and deposit their energy within a particular layer of the camera.

As mentioned in the introduction, the circular structures in the ion pinhole images are not unique to these experiments and have been observed in DPF since 1980 [18, 19, 20–25]. However, the experimental results obtained in those experiments did not allow researchers to unambiguously determine whether the ring-shaped patterns reflected the real spatial distribution of the ion source or whether they were created by magnetic fields focusing ions from one divergent point source.

In our opinion, it was necessary to obtain additional experimental data to elucidate the origin of the patterns recorded by the ion pinhole cameras. In working towards that goal, we used various ion detectors. Results relevant to

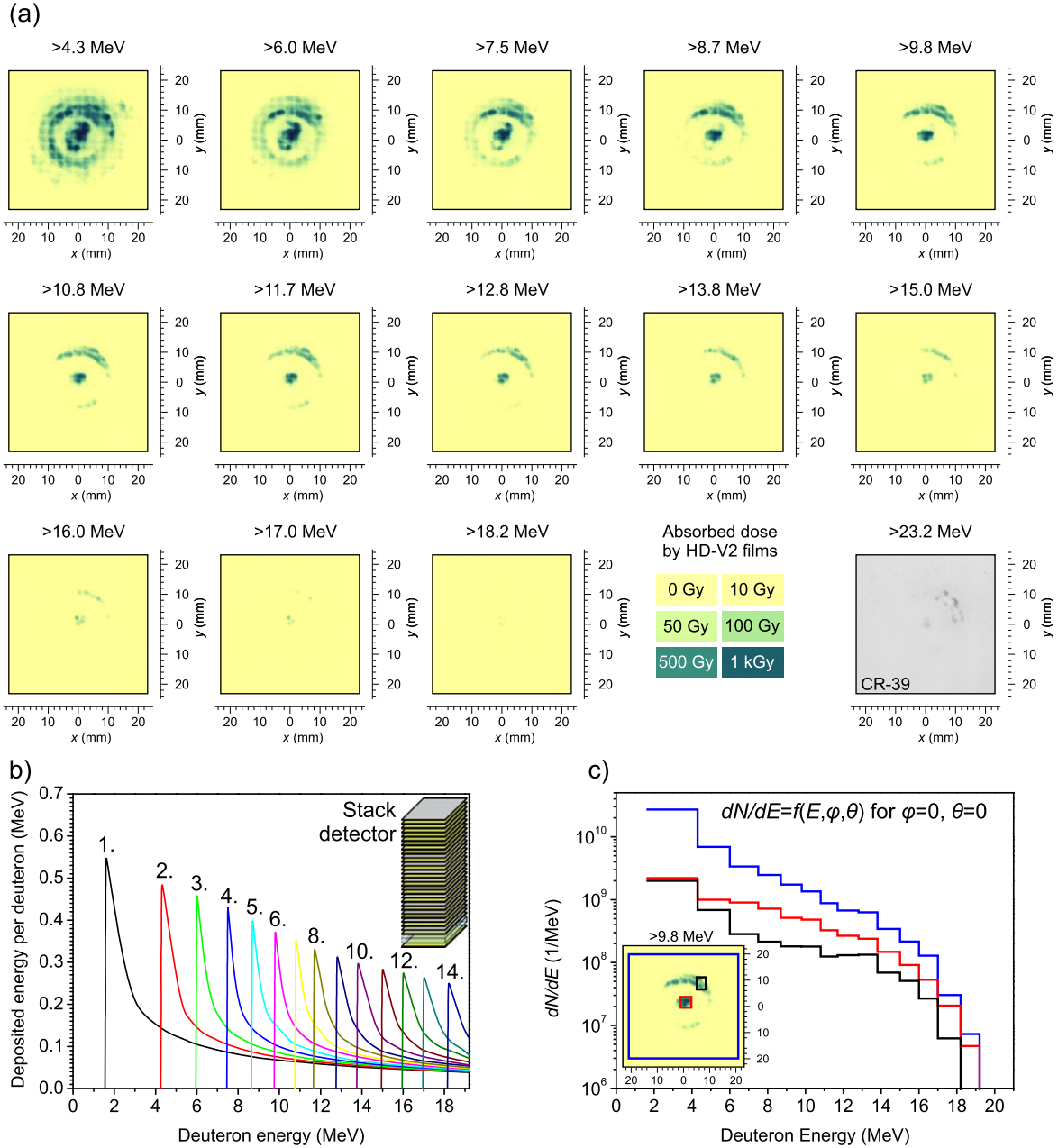


Figure 4. (a) Images from the ion pinhole camera recorded by the stack of HD-V2 films and a CR-39 detector with a spatial resolution of 1.2 mm. The energy refers to deuterons. The contribution of protons to RCF doses is expected to be significant mainly at the highest energies. (b) Spectral response of RCFs in the stack calculated with the SRIM/TRIM code [39]. (c) Deuteron energy distribution functions unfolded from the stack of RCF films. Three spectra (black, red, and blue) correspond to three regions selected in the ion pinhole image at a >9.8 MeV deuteron energy. Shot no. 1771 with $(2.0 \pm 0.4) \times 10^{12}$ neutrons.

understanding the structure in the ion pinhole images will be presented in the following sections 4–6.

4. Ion pinhole images with spectral resolution

In order to obtain additional information about the patterns in the ion pinhole camera images, we took advantage of the multi-MeV ion energies produced on GIT-12. High-energy deuterons generated in our experiment enabled us to record images on multiple detectors placed in the stack of the pinhole camera. If there is a sufficient number of detectors in the

stack, as in the shot presented in figure 4(a), it is possible to unfold the depth-dose distribution to obtain a deuteron spectrum through so-called RCF stack spectroscopy [31, 32]. This method is based on the fact that each RCF in the stack has a different sensitivity to deuterons (see figure 4(b)). We used a numerical code to process the images, and the unfolded spectra are shown in figure 4(c). The spectrum obtained depended on the region selected for analysis within the exposed area of the detector. The spectrum from a small part of the ring is plotted in black. This region is highlighted by the black square in the accompanying detector image. A significant result is that this ring of nearly constant diameter

through the film stack was formed by ions with various energies ranging from ~ 1 MeV up to 20 MeV. Consequently, we can rule out the hypothesis from [18] that the ring was produced by magnetic fields focusing ions from one divergent central point source towards a pinhole. In the case of a divergent point source, the ring diameter in pinhole images would strongly depend on the ion energy, since the Lorentz force depends on the deuteron velocity. To produce a nearly constant diameter ring spanning such a wide range of energies, ions of a specific energy would require either that their trajectory be bent by a specific magnetic field or that they be accelerated at a specific initial angle with respect to the axis. It is not likely that such specific conditions reliably occurred over so many shots with different initial parameters during various experimental campaigns. Therefore, in our previous paper [26], we arrived at the conclusion that the ring in the pinhole images somehow reflected the spatial distribution of the ion source. The relationship between the diameter of the circular source at its origin and that observed in the plane of the cathode mesh depends on the parameters of the ion source above the cathode mesh and on the strength of the electric and magnetic fields inside the z-pinch. Taking into account the complexity of the electric and magnetic fields during the ion acceleration in z-pinch, we admit that there are other possible explanations for the formation of a nearly constant diameter ring by ions of various energies. We are not able to make definite conclusions based on images recorded by only one ion pinhole camera with spectral resolution. Therefore we present additional results in the following sections 5 and 6.

5. Ion multi-pinhole images

It has been shown in the previous section that the annuli observed by the ion pinhole camera somehow represent the real spatial distribution of the ion source. To study the real diameter and other parameters of the ion source at its origin, it seems desirable to detect ions using many pinholes distributed at different distances from the cathode mesh that observe the z-pinch along different lines of sight. For this purpose, a multi-pinhole camera was added to the top of the detector shown in figure 3. This addition allowed us to obtain results from one on-axis and four off-axis pinhole cameras (see figures 5(a)–(c)). The absorber and the first five layers in the stack of the off-axis pinhole cameras were the same as the layers in the on-axis camera presented in figure 4. The series of pinhole images at >4.3 MeV and >7.5 MeV from shot no. 1771 (i.e. the same shot as presented in figures 2 and 4) are shown in figures 5(b)–(c).

A significant result is that, at the 4.3 MeV deuteron energy, the ring was observed by all pinhole cameras regardless of the line of sight and the distance of the pinholes from the cathode mesh. It supports the hypothesis that the ring was real. However, it does not mean that magnetic fields played no role. The magnetic field likely manifested itself in a blurring of the concentric ring at lower ion energies (see figure 5(b)) and in the ion emission anisotropy observed by the pinhole camera along different lines of sight at higher energies (see differences between images in figure 5(c)).

To find out the real diameter of the ion source, we tried to reproduce the most important features observed in the multi-pinhole images by numerical simulations. Our numerical simulation did not address the acceleration of the ions. In our charged-particle tracking code, we assumed that ions were already accelerated to their final energy. Then we simulated trajectories of these ions in azimuthal magnetic fields generated by the z-pinch current. By comparing simulated and experimental multi-pinhole images at many ion energies, it was possible to study both the real diameter of the ion source and the influence of magnetic fields on ions detected by the multi-pinhole camera. The geometry of the numerical simulation and the simulated pinhole images are shown in figures 5(d)–(f). The set-up of the simulations was as follows. In our simulations, we placed the source of accelerated ions 18 mm above the cathode mesh. The anode–cathode gap was 25 mm. The ion source consisted of two parts, namely a central spot and a ring. As shown in figure 5(d), the free parameters of our simulations were the radius of the ring-shaped source, the (energy-dependent) angular divergence of the ion micro-beams from the z-axis, and the current at a specified radius (skin or uniform current density). The azimuthal asymmetry was reproduced by the variable intensity of the ion source along the ring and by a slight (~ 1 mm) displacement of the multi-pinhole camera from the z-axis. The pinhole diameters and the geometry of the pinhole camera were the same as in the experiment. The quantitative agreement of the numerical simulation with our experiment was obtained with the skin current distributed at a 9 mm radius. The current was time-varying and was decreasing from a relatively low peak value of ~ 1 MA. However, we should note that this value would be higher if the source were placed at a lower height above the cathode mesh or if the initial directions of the ion micro-beams were not oriented along the z-axis. Despite this ambiguity, the initial radius of the ring-shaped source was (10 ± 1) mm in all our simulations of shot no. 1771. By analyzing other shots, we estimated the initial radius of the ion source to be between 6 and 10 mm.

The radius of the ring-shaped ion source is not the only important result encoded in the ion pinhole images. The multi-pinhole camera could also provide information about the ion-beam divergence. At deuteron energies above 5 MeV, the characteristic feature was that each ion pinhole camera detected mainly those localized spots which lie in the plane determined by the z-pinch axis and that particular pinhole (see figure 5(f)). This observation can be explained by localized sources and collimated ion emission which is deflected by azimuthal magnetic (or radial electric fields) as shown in figure 5(d) (see also figure 6 in [26]). This forms radial lines at the beam-profile detector (see figure 7(b)). Each individual source can be detected by a pinhole camera if the radial line hits a pinhole (see figure 6 in [26]). As a result, the central (on-axis) pinhole near the center of the beam (the cross-section of many of the radial lines) sees the largest number of localized ion sources. The off-axis pinholes can detect sources which lie in the plane determined by the pinhole and the z-pinch axis. Other sources can be seen only if the initial micro-beam divergence is sufficiently large. This caused us to consider investigating the ion

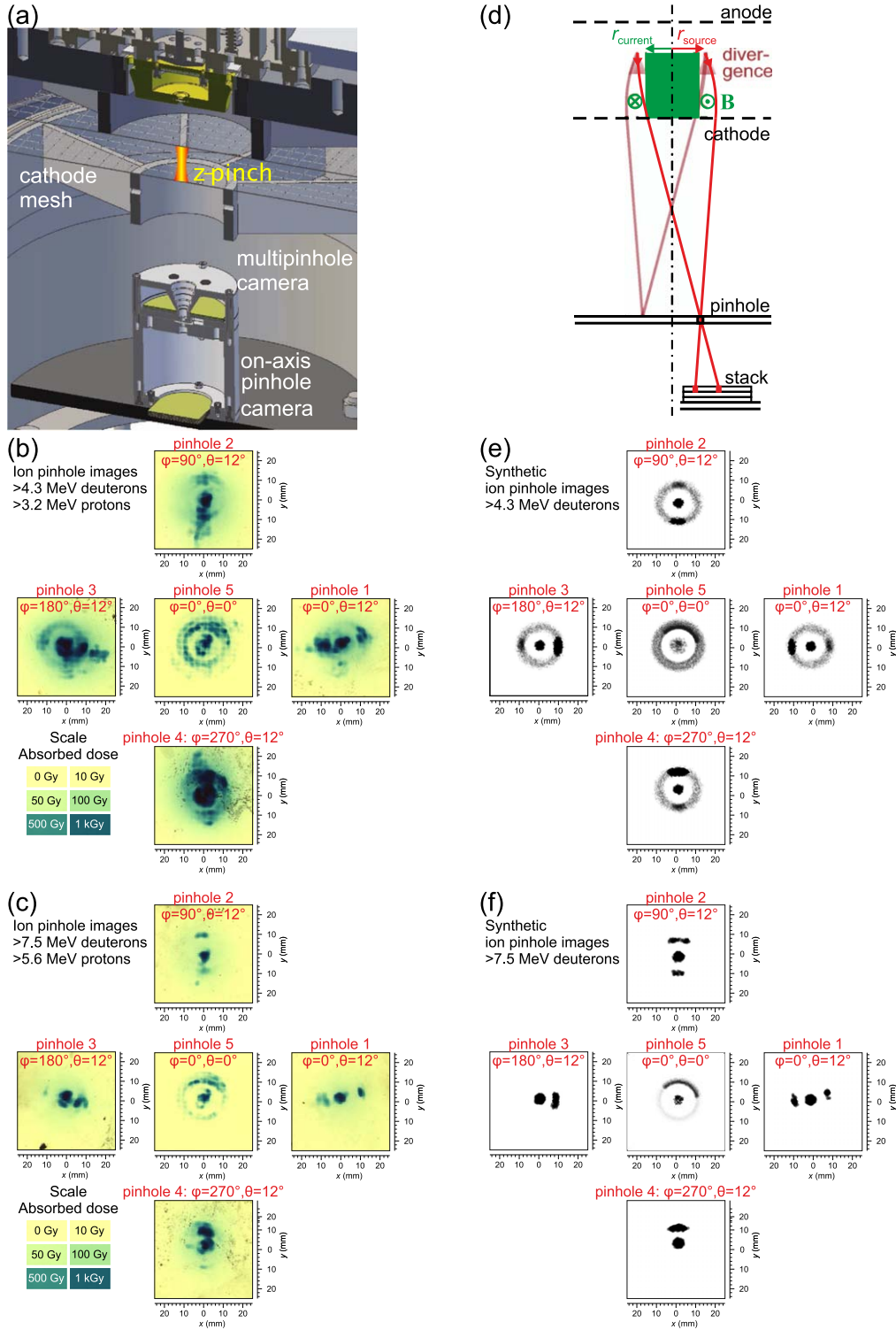


Figure 5. Measurement of the spatial distribution of the ion source. (a) Arrangement of the ion multi-pinhole camera with pinhole diameters of 0.5 mm, magnifications of 0.6 and 0.3, and spatial resolutions of 1.3 and 2.2 mm for the on-axis and off-axis pinholes, respectively. (b)–(c) Experimental (one on-axis and four off-axis) ion pinhole images at 4.3 and 7.5 MeV deuteron energies on shot no. 1771 with $(2.0 \pm 0.4) \times 10^{12}$ neutrons. Spatial scales correspond to the plane of the cathode mesh. (d) Geometry used for the simulation of deuterons detected by the multi-pinhole camera. For simplicity, the ion emission from the central spot is not shown. Not to scale. (e)–(f) Synthetic ion pinhole images at 4.3 and 7.5 MeV deuteron energies.

micro-beam divergence using off-axis pinhole images. For instance, at an 7.5 MeV energy, the synthetic images in figure 5(f) were simulated with ions emitted into 100 mrad and

250 mrad cones for the ring source and the central spot, respectively. The ion-beam divergence measured in a wider energy range will be presented in the following section 6.

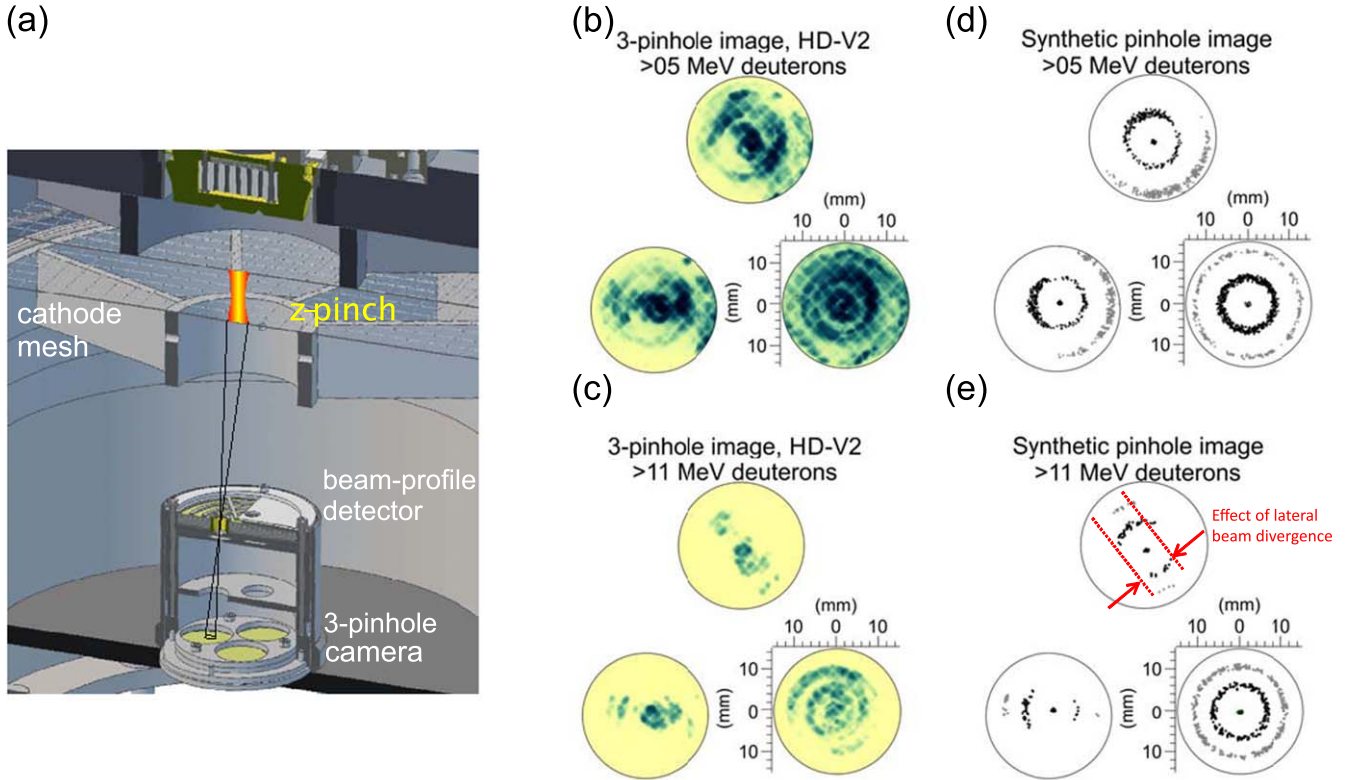


Figure 6. Measurement of the micro-beam divergence from three-pinhole images. (a) Arrangement of the ion three-pinhole detector with pinhole diameters of 0.45 mm, a magnification of 0.55, and a spatial resolution of 1.3 mm. (b)–(c) Experimental three pinhole images at 5 and 11 MeV deuteron energies on shot no. 1845 with $(3.4 \pm 0.6) \times 10^{12}$ neutrons. (d)–(e) Synthetic ion pinhole images at 5 and 11 MeV deuteron energies. Spatial scales correspond to the plane of the cathode mesh. Note: The pinhole images showed two ring-like structures. The radius of the outer ring seemed to depend on the deuteron energy. Therefore we will try to verify the hypothesis that both rings came from one original ring-shaped source in future simulations. Regardless of their origin, two rings helped us to measure the beam divergence with greater precision.

6. Ion three-pinhole images and ion beam-profile

We suggested in the previous section 5 that micro-beam divergence could be estimated from off-axis ion pinhole images. For this purpose, we used a three-pinhole camera with a better spatial and spectral resolution. The detector is shown in figure 6(a) and described in [26]. This detector measured angular, spatial, and spectral properties of ion emission. The angular distribution (ion-beam profile) was measured by using large pieces of ion detector materials (CR-39, GafChromic films) at about 10 cm below the cathode mesh. The effective area of the detectors was reduced by a shielding mask and 3 cutouts. At the center of each cutout, there was a pinhole, thereby providing the spatial distribution of the ion source. The pinhole diameters were chosen between 0.25 and 0.45 mm. Calculating with the image magnification of 0.55, the pinhole camera provided a spatial resolution between 0.7 and 1.25 mm. Three pinholes were used to estimate the anisotropy of the ion emission and consequently the ion microbeam divergence. On shot no. 1845 presented in figure 6, we used a stack of 21 GafChromic films (see also figure 5(a) in [26]). Figures 6(b) and (c) show typical pinhole images of deuterons with energies above 5 and 11 MeV, respectively. In the images, it is evident that the deuterons produced a radiograph of the cathode mesh. The marks in the cathode mesh (spatial fiducials) enabled us to compare the absolute radial position of the ion

emission seen along different lines of sight, i.e. seen by different off-axis pinholes. At low energies, the rings were blurred and visible in all images of the three-pinhole camera. Above 5 MeV, we can see differences between individual pinholes (see figure 6). Above 11 MeV, the whole ring was detected only in the bottom-right pinhole image which corresponded most closely to the center of the ion beam. The off-center pinhole images recorded only those parts of the rings which laid in the plane determined by the beam axis and a pinhole.

We tried to reproduce the characteristic features mentioned above in simulations. In a manner similar to the simulation in figure 5, the radius of the ring-shaped source, the current at a specified radius, and the divergence of the localized ion sources were free parameters whose values were chosen for the simulation. The divergence of ions emitted from the ring is connected to the width of arcs observed in off-center pinhole images. This width is marked by a red color in figure 6(e). The width and, therefore, the reconstructed ion micro-beam divergence decreased with increasing deuteron energy. The result obtained on shot no. 1845 is shown in figure 7(a). At 5 MeV, the lateral micro-beam divergence was 140 mrad whereas it decreased to 10 mrad at 26 MeV. The latter value is in agreement with the result from the beam-profile detector. At deuteron energies above 18 MeV, the beam-profile detector recorded distinguishable radial lines. The width of the radial stripes in figure 7(b)

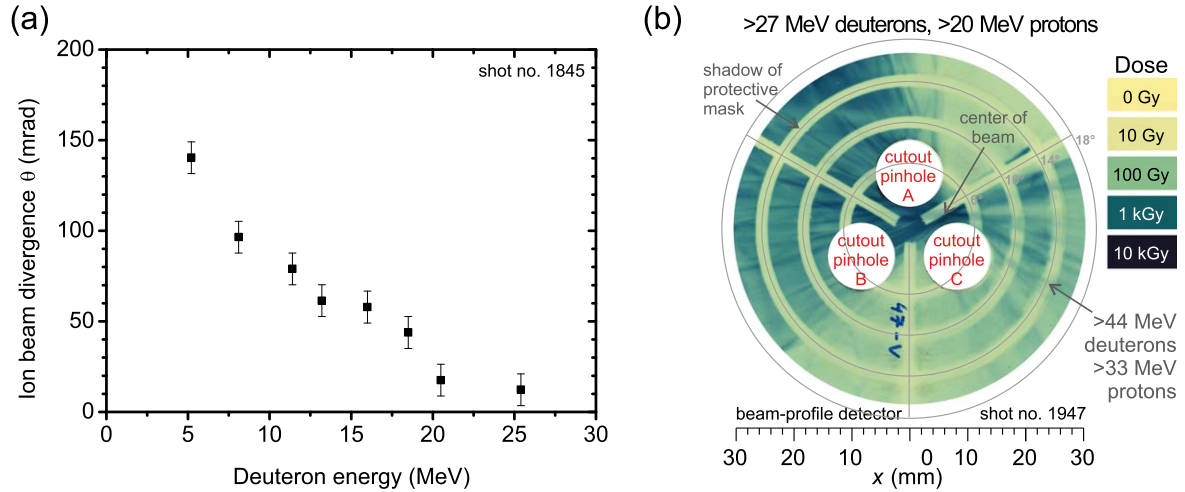


Figure 7. (a) Dependence of the ion-beam microdivergence on deuteron energy, estimated for shot no. 1845. (b) An image of the beam profile detected by HD-V2 GafChromic film at 10 cm on shot no. 1947.

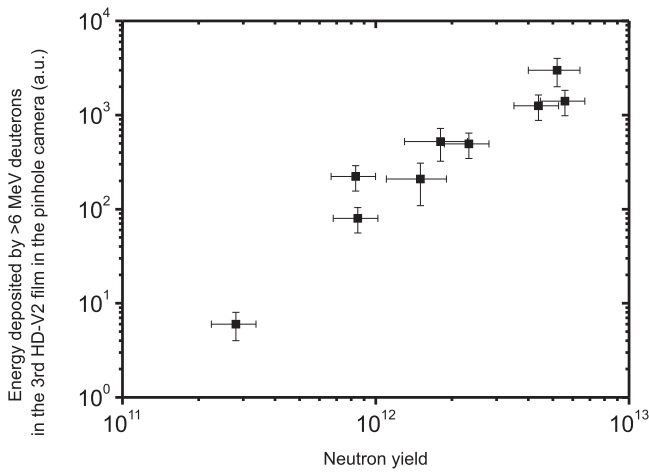


Figure 8. Correlation between the neutron yield and the energy deposited by >6 MeV deuterons in the third HD-V2 film in the on-axis pinhole camera. We intentionally selected shots with different neutron yields observed in 2016. The differences in neutron yields were caused by shot-to-shot variation or by non-ideal initial parameters (on shots with different aims than a high-neutron yield or on shots before the optimal conditions were set up).

implies a lateral divergence between 5 and 10 mrad at 27 MeV deuteron energy. This demonstrates the consistency of the pinhole images and the beam-profile patterns. In the next section, we will present the correlation of pinhole images with neutron yields.

7. Ion pinhole images and neutron yields

Deuterium z-pinch and DPF are researched primarily as neutron sources. From this point of view, it is interesting to determine if there is any correlation between the neutron yield and features of the ion pinhole images. In our experiments on GIT-12, the most obvious correlation was seen between the neutron yields and the dose absorbed in the GafChromic films fielded in our pinhole cameras. For example, figure 8 presents

the neutron-yield dependence of the energy deposited by >6 MeV deuterons in the GafChromic films placed in the on-axis pinhole camera.

At deuteron energies below 5 MeV, the HD-V2 GafChromic films were often saturated. In figure 8, we therefore show the doses produced by >6 MeV deuterons which correlated with neutron yields quite well. If the shape of the energy spectrum of fast deuterons is approximately the same from shot to shot, then the relative neutron yield from $^2\text{H}(d, n)^3\text{He}$ fusion reactions is given by the numbers of accelerated and target deuterons. Therefore, the close connection of neutron yields with the doses produced by accelerated deuterons is not surprising.

We also looked for a correlation between the neutron yields and ring diameters but did not find any. In the following section 8, we will therefore present another attempt to identify the origin of a ring with a particular diameter.

8. Origin of the ring ion source

8.1. Anode pinhole camera

Various research groups have measured ring-like structures with ion pinhole cameras in DPF and z-pinch experiments since 1980. In our experiments on GIT-12, we also observed annuli along with the expected on-axis spots in pinhole camera images produced by multi-MeV deuterons. In the previous sections 4 and 5, we showed that the ring represented the real spatial distribution of an ion source with a radius of about 1 cm. This is an extremely important result to include in discussion of the ion acceleration mechanism (see e.g. [26]). Since the estimated diameter of the ion source was significantly larger than the diameter of the compressed necks observed shortly before the acceleration of the ions (see figure 2(b)), it is natural to ask what sets the diameter of the ring.

To explain a particular diameter in our experiment, we looked for any correlation between the estimated diameter of the ring and results from other diagnostics. We found out that 15–20 mm diameters were characteristic for the plasma

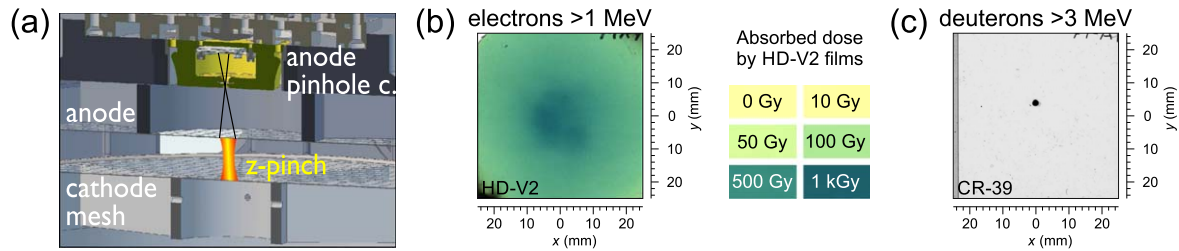


Figure 9. (a) Arrangement of the anode pinhole camera with a magnification of 0.55, a pinhole diameter of 0.6 mm, and a spatial resolution of 1.7 mm. (b) Anode pinhole image recorded by the third HD-V2 GafChromic film in the stack. (c) Pinhole image detected by the first CR-39 detector. Shot no. 1771. The stack consisted of a 50 μm aluminum absorber, a 0.65 mm CR-39 detector, the first HD-V2 film, a 0.55 mm CR-39 detector, the second HD-V2 film, a 1 mm aluminum absorber, the third HD-V2 film. Spatial scales correspond to the plane of the anode mesh.

emission near the anode (observed by a time-integrated x-ray pinhole camera as shown in figure 2(c)). We hypothesized that the plasma emission at the anode mesh could be a consequence of electron flow at stagnation or disruption. That is why we were interested in the spatial distribution of the electron beam at the anode. For this purpose, we decided to make an ‘anode’ pinhole camera inside the gas-puff hardware.

The arrangement of the anode pinhole camera is displayed in figure 9(a). Figures 9(b) and (c) then show the results obtained on shot no. 1771 (the same shot as presented in figures 2, 4, and 5). The electron beam was detected by the stack of HD-V2 GafChromic films. The third HD-V2 film in the stack recorded the pattern shown in figure 9(b), which is a circle that is not hollow. The stopping range of electrons in our stack was calculated by the ESTAR program provided by the National Institute of Standards and Technology (NIST) [42].

HD-V2 GafChromic films are sensitive also to ions and x-rays. The contribution of ions was studied by the CR-39 detector which was placed in front of the HD-V2 films. On shot no. 1771, the CR-39 detector behind a 50 μm Al absorber recorded only a small spot which was created by protons or deuterons with energies above 2.3 MeV or 3.0 MeV, respectively (see figure 9(c)). The etching and the visibility of tracks could be influenced by high dose x-ray and electron irradiation [43–45]. Nevertheless, it was clear that the total number of $>2\text{MeV}$ hydrogen ions emitted towards the anode was at least three orders of magnitude lower than towards the cathode. This is in agreement with the previous ‘upstream’ measurements in DPFs [23]. We suppose that some deuterons could be accelerated in the direction of an electron beam by a collective ion acceleration mechanism [46]. Another hypothesis is that fast protons from $^2\text{H(d,p)}^3\text{H}$ fusion reactions escaped the z-pinch towards the anode along the z-axis. To estimate the number of $>8\text{MeV}$ hydrogen ions emitted towards the anode, we analyzed the rear side of the first CR-39 detector, which had a thickness of 0.65 mm. Even though we observed a few proton or deuteron tracks above the background (neutron) level, the contribution of ions to the dose detected by the HD-V2 films behind the CR-39 detector can be neglected. As far as the contribution of hard x-rays is concerned, the patterns in the HD-V2 films were shifted when magnets were placed inside the pinhole camera. We may,

therefore, conclude that the circular pattern of about 15–20 mm diameter in figure 9(b) was produced by fast electrons with energies above 1 MeV. It should be noted that the recorded image could be influenced by the transport of the high-current electron beam in the low-pressure gas (or plasma) between the anode mesh and the pinhole camera. Based on the shadow of the anode mesh in the pinhole images, the original electron beam diameter at the anode mesh could be 1.6 times smaller than the one shown in figure 9(b). Besides that, even if the recorded pattern is of a size similar to that of the rings in the ion pinhole images, it does not explain why the ion pinhole camera below the cathode mesh detected only individual annuli of a specific diameter. Consequently, we have to search for the origin of a particular diameter in a different way.

Another possible way to study the origin of the annuli in the (cathode) ion pinhole images is to change the initial parameters of the GIT-12 experiments. In previous experimental campaigns, we changed the length of the anode–cathode gap, or we placed cylindrical objects of various diameters on the anode or cathode meshes. We also tried to influence the ion emission by making a hole in the meshes. We believed that the rings in the pinhole images would be changed and we would see some correlation with the initial electrode configuration. Unfortunately different initial parameters usually had deleterious effects on the ion acceleration and neutron production. Therefore, the results from these experiments have not been conclusive so far. To study how the diameter of the ring in the ion pinhole images might change, we instead decided to make measurements with the ion pinhole camera on a different pulsed-power generator. Recently, we have had an opportunity to carry out dense-plasma focus experiments on the HAWK pulsed-power generator. The results from our ion pinhole camera are presented in the following subsection.

8.2. Ion pinhole images on HAWK at the Naval Research Laboratory

The experimental arrangement on HAWK is displayed in figure 10(a). Figure 10(b) shows an on-axis view with the valve inserted inside the central anode. Our ion detector was placed 20 cm from the anode. We used the same three-pinhole camera as on GIT-12 (see figure 6). With an image

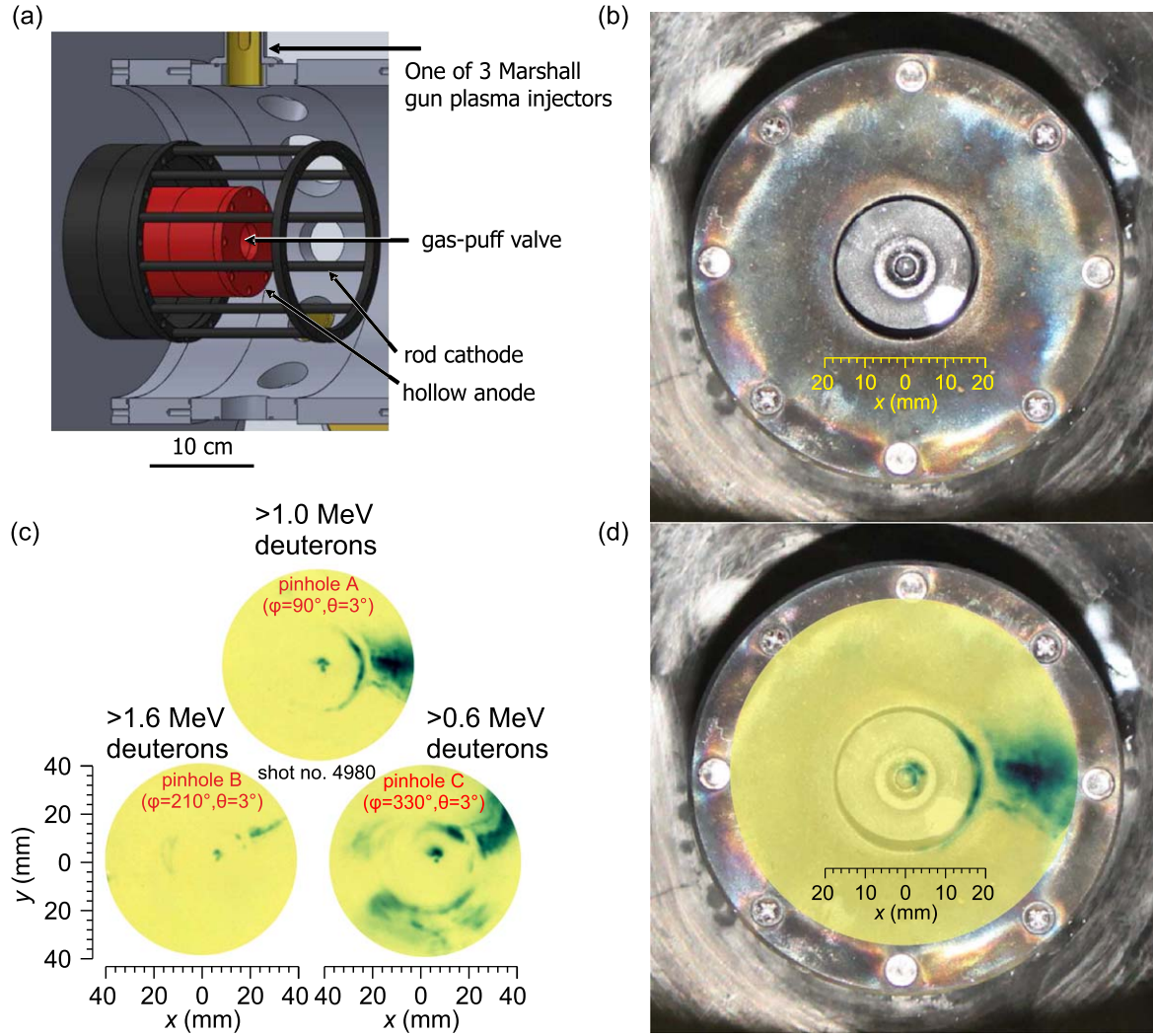


Figure 10. (a) Arrangement of the electrode system on the HAWK generator. (b) On-axis view of the anode with the gas-puff valve inside the central conductor. (c) Images from the three-pinhole camera (with magnification of 0.25, pinhole diameter of 0.4 mm, spatial resolution of 2 mm) recorded by the HD-V2 film behind three different absorbers ($6\ \mu\text{m}$ Kimfoil, $10\ \mu\text{m}$ Al, and $20\ \mu\text{m}$ Al). (d) On-axis view of the anode overlapped by the pinhole image at the >1.0 MeV deuteron energy. All spatial scales correspond to the plane of the anode end.

magnification of 0.25 and a pinhole diameter of 0.4 mm, the pinhole camera provided a spatial resolution of 2 mm. The detector stack consisted of a layer of HD-V2 GafChromic film followed by two EBT-3 films. In front of three individual parts of the first HD-V2 film, we placed three different absorbers, namely $6\ \mu\text{m}$ Kimfoil, $10\ \mu\text{m}$ Al, and $20\ \mu\text{m}$ Al. Pinhole images from a typical shot are shown in figure 10(c). Despite the 0.64 MV output voltage and relatively modest current of the HAWK generator, we recorded a high dose behind all three absorbers. The presence of signal behind the $20\ \mu\text{m}$ Al absorber implies deuteron energies above 1.6 MeV. Further, in figure 10(c) we can see that the diameter of the characteristic rings is the same in the three pinhole images, despite the fact that each of these images corresponds to a different deuteron energy and line of sight. As on GIT-12, we assume that the electric and magnetic fields inside the plasma could influence the trajectory of ions traveling between the anode and the pinholes. However, an important result that is apparent from figure 10(d) is that for HAWK, the diameter

of the rings in all of the pinhole images matches the diameter of the gas-puff hole inside the anode. This result indicates that the spatial distribution of the ion emission could be influenced by the geometry of the electrodes. Several implications of these findings will be discussed in the last section 9.

9. Discussion and conclusions

Mega-ampere DPF and deuterium gas-puff z-pinches can produce high-energy ions. Recently, we have accelerated deuterons to multi-MeV energies in deuterium gas-puff experiments on the GIT-12 and HAWK microsecond generators at 3 MA and 0.7 MA current, respectively. We used a large number of different detectors to characterize the ion emission and determine specific details about the nature of the ion source and acceleration mechanism. In particular, the results from ion pinhole cameras fielded during the

experiments seemed to be useful for the discussion of ion acceleration mechanisms.

In the ion pinhole images on GIT-12, we observed rings of nearly constant diameter at various deuteron energies ranging from ~ 1 MeV up to 20 MeV. Despite some anisotropy in the ion emission, these rings were observed by cameras with pinholes which were distributed at different distances from the z-pinch and which observed the z-pinch along different lines of sight. Based on measurements with one-, three-, and multi-pinhole cameras, we therefore concluded that the observed rings reflected both the spatial distribution of the ion source and the influence of magnetic fields. The estimated ion source diameter on the order of 1 cm was significantly larger than the width of the necks in the plasma observed before the disruption of the current (see figure 2(b)). This knowledge helped us to determine specific details about the acceleration mechanism. These details are consistent with the mechanism that we described in [26] which is based on the generation of GV m^{-1} electric fields during fast ($\sim \text{ns}$) disruption of the conduction current (see also [47, 48]). A significant current drop in our experiments is attributed to a transition from a low-impedance plasma to a space-charge-limited flow and/or self-magnetic insulation in a gap formed after the ejection of plasmas from $m = 0$ constrictions. Then the electric field is induced in regions where the conduction current was flowing before the disruption. We do not expect that the total current flows within the ~ 1 mm neck observed in framing camera images from GIT-12 (see also [49]). In addition to the current flowing outside the observed neck, the generated electromagnetic pulse can propagate throughout the whole inter-electrode region. Therefore, it is not surprising that ions were accelerated from the anode (plasma) region not only on the axis but also at a larger (~ 1 cm) diameter. We still need to explain why the ring-shaped portion of the ion source forms at a particular diameter. We hypothesize that the current density distribution plays a role in determining this diameter, but we do not know the current density distribution in the z-pinch on GIT-12 before the disruption.

A significant result explaining why a ring-shaped ion source might form at a particular diameter was achieved on the HAWK generator at the US Naval Research Laboratory in Washington. Despite differences in initial conditions, the experiment on HAWK exhibited many similarities to GIT-12. When the three-pinhole camera was placed on the axis, we saw characteristic rings behind different absorbers and along different lines of sight. The apparent correlation of the ion-ring diameter with the diameter of the gas-puff hole inside the anode demonstrated the importance of the electrode edge on the ion emission. The influence of the electrode geometry on the ion acceleration was also clearly visible in DPF experiments on PF-1000 [24]. Taking all the above considerations into account, the spatial profile of the original ion source seems to be determined both by the current density distribution before the disruption and by the geometry of the experiment. Then the properties of the ion emission are determined by the (secondary) breakdown in the anode–

cathode gap when the unstable plasma is unable to carry the total conduction current.

So far, our conclusions about the ion pinhole images have been drawn from experimental observations. It would be desirable to support our conclusions about the ion acceleration mechanism by reproducing the experimental results with state-of-the-art numerical codes. Unfortunately, we have not found any numerical code capable of simulating ion acceleration in z-pinches that could explain the experimentally-observed ion pinhole images. When a numerical code is developed that is able to simulate the most important experimental details, it will then be reasonable to move from researching the acceleration mechanism to learning how to intentionally influence the ion acceleration.

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